

Design and Implementation of a Compiler with Machine-Learning-Guided Optimization Selection

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Abstract—This paper documents the complete design, implementation, and evaluation of MiniLang, a research compiler for a minimal statically typed imperative language. MiniLang was build from scratch in C++17 to support end-to-end study of compiler construction and machine-learning-guided optimization. The system implements a classical multi-phase pipeline: hand-written lexical analysis, recursive-descent parsing, two-pass semantic analysis, lowering to a flat High-level Intermediate Representation (HIR), configurable rule-based optimizations, textual LLVM IR emission, and an ML-ready optimization advisor with evaluation harness.

We describe every compilation phase in implementation-level detail, including token design, abstract syntax tree (AST) structure, semantic rules, HIR instruction set, three optimization passes, LLVM code generation strategy, and a feature-driven advisor that selects optimization pipeline using heuristics, JSON plans or external Python interface. Experimental results in benchmark programs show that optimization reduces the HIR instruction count by up to 63% ($22 \rightarrow 8$ instruction) and LLVM IR size by 41% ($34 \rightarrow 20$ lines) on constant-heavy workloads, while preserving correct runtime behavior (verified by process exit codes.) Minilang is intended as a reproducible research and teaching traditional compiler engineering and modern ML-for-compilers research

Index Terms—Compilers, intermediate representation, LLVM, static analysis, code optimization, machine learning, pass ordering

I. INTRODUCTION

A. Motivation

Optimizing compilers transform source programs through dozens of analysis and transformation passes. Deciding which passes to run, and in what order has been tuned manually for decades in systems such as GCC and LLVM[1],[2]. Recently, machine learning (ML) has been applied to predict profitable compiler decisions—including pass ordering, heuristics flags, and inlining thresholds—from program features[3]–[6]. However, experimenting with ML-guided optimization typically requires navigating industrial-strength compiler infrastructures whose complexity obscures the relationship between front-end structure, IR design, and optimization outcomes.

Minilang addresses this by providing a complete but minimal compiler where every phase is inspectable, independently executable, and instrumented for measurement. This project follows a staged research methodology:

- Language design - define a small typed language sufficient for control flow, functions, and arithmetic.

- Front-end construction- implement lexer, parser, and semantic analyzer with explicit error reporting.
- IR design - lower AST to flat HIR with explicit temporaries and structured control flow.
- Rule-based optimization - implement classic local optimizations at HIR level.
- Code generation - emit human-readable LLVM IR for execution via Clang.
- ML advisor integration - extract static features, select pass pipelines, and evaluate against baselines.

B. Research Questions

This work investigates:

- RQ1: Can a minimal imperative language support a full, pedagogically transparent compilation pipeline comparable to textbook architecture [1],[7]?
- RQ2: Is a flat HIR with explicit temporaries a suitable layer for both rule-based optimization and feature extraction?
- RQ3: Can static HIR metrics drive selective application of optimization passes with measurable IR reduction?

C. Contributions

- Formal specification of the MiniLang language (lexical, syntactic, and semantic rules).
- Complete architectural documentation of a C++17 compiler with six standalone driver tools.
- HIR specification with lowering algorithms for expressions, conditionals, and loops.
- Three HIR optimizations with fixed-point iteration and statistics.
- ML-ready advisor framework: 14-dimensional feature vector, pass, registry, heuristic/JSON/Python modes, and minilang_eval harness.
- Empirical evaluation demonstrating IR reduction and correctness on benchmark programs.

D. Paper Organization

Section II reviews related work. Section III defines MiniLang. Section IV presents system architecture. Sections V–X detail each compilation phase. Section XI covers the ML advisor. Section XII presents evaluation. Section XIII–XIV discuss limitations and conclude.

II. RELATED WORK

A. Compiler Construction Foundations

The classic compiler pipeline- lexical analysis, syntax analysis, semantic analysis, intermediate code generation, optimization and code generation- is established in Aho et al.[1] and Cooper and Torczon [2]. MiniLang adopts this structure verbatim, using a hand-written scanner and recursive-descent parser rather than generator tools(Flex/Bison) to maximize transparency for readers and students.

Appel [8] and Muchnick [9] emphasize the role of intermediate representation in decoupling front-end language features from back-end code generation. Our HIR follows the principle: it is lower than AST(flat instruction list) but higher than LLVM (no SSA no target- specific details).

B. LLVM as a Back-end

Lattner and Adve introduced LLVM as a modular compilation framework with SSA-based intermediate representation and aggressive optimization infrastructure [2]. Rather than linking against the LLVM C++ API initially, MiniLang emits textual LLVM IR (.ll files), which Clang consumes for native code generation. This approach mirrors the LLVM Kaleidoscope tutorial [10] and keeps the research artifact readable in published papers.

C. Machine Learning for Compiler Optimization

Ashouri et al. provide a comprehensive survey of ML applied to compiler autotuning, pass ordering, and heuristic selection [3]. CompilerGym exposes LLVM optimization as a reinforcement-learning environment [4]. Other work learns graph representations of programs for heuristic tuning [5],[6]. MiniLang differs from these systems in scope and interpretability: optimization decisions occur at HIR level with explicit, low-dimensional features (14 integers), making it suitable for classroom demonstration and lightweight ML prototyping without requiring GPU-scale training infrastructure. The advisor is architected as a pluggable policy (heuristic today; ONNX or Python model tomorrow), aligning with survey recommendations for separating feature extraction from inference [3].

D. Educational Compilers

Educational compilers traditionally prioritize clarity over performance. MiniLang extends the educational pattern by adding (1) a documented semantic analyzer with symbol tables (2) a serialized HIR dump format, and (3) quantitative evaluation tooling- bridging teaching and research.

III. THE MINILANG PROGRAMMING LANGUAGE

MiniLang is intentionally small. It supports integer and boolean computation, functions conditionals, and loops. but omits pointers, arrays, heap allocation, and I/O builtins.

A. Lexical Structure

Category	Rules
Whitespace	Space, tab, newline; tracked for source locations
Comments	// to EOL; /* ... */ block comments
Identifiers	[A-Za-z_][A-Za-z0-9_]*
Integers	[0-9]+ (decimal, signed via unary -)
Keywords	int, bool, void, if, else, while, return, true, false
Operators	+ - * / %, comparisons, && ! =
Delimiters	() { } [] ; ,

TABLE I: MiniLang Lexical Rules

The lexer classifies keywords via `classify_keyword()` after scanning identifiers. Multi-character operators(==, !=, ≤, ≥, &&, ||) are recognized using one-character lookahead (`match()`).

B. Context-Free Grammar

MiniLang grammar is LL(1)-compatible:

program	::= function*
function	::= type IDENT "(" [parameters] ")" block
parameters	::= type IDENT ("," type IDENT)*
block	::= "{" statement* "}"
statement	::= block type IDENT "=" expression ";" "if" "(" expression ")" statement ["else" statement] "while" "(" expression ")" statement "return" [expression] ";" expression ";"
expression	::= logical_or
logical_or	::= logical_and (" " logical_and)*
logical_and	::= equality ("&&" equality)*
equality	::= comparison (("==" "!=") comparison) *
comparison	::= term (("<" "<=" ">" ">=") term)*
term	::= factor (("+" "-") factor)*
factor	::= unary (("*" "/" "%") unary)*

unary	::= ("!" "-") unary call
call	::= primary
primary	::= INT "true" "false" IDENT ["(" [arguments] ")" "(" expression ")"
arguments	::= expression ("," expression)*
type	::= "int" "bool" "void"

Operator precedence increases toward primary:logical OR (lowest) through multiplicative operators (highest). Unary operators bind tighter than binary operators.

C. Abstract Syntax Tree

The AST uses a tagged-union style with explicit Kind enumerations:

- Expressions: IntLiteral, BoolLiteral, Variable, Binary, Unary, Call.
- Statements: Block, VarDecl, If, While, Return, Expr
- Top level: (*Program* $\rightarrow$ *vector*) of FunctionDecl

Each node carries a SourceLocation (filename, line, column) for diagnostics. Types are represented by TypeKind: Int, Bool, Void.

Example. The factorial program:

```
int factorial(int n) {
    if (n <= 1) {
        return 1;
    }
    return n * factorial(n - 1);
}

int main() {
    int x = factorial(5);
    return x;
}
```

parses into two FunctionDecl nodes with nested IfStmt, ReturnStmt, CallExpr. and VarDeclStmt nodes.

D. Static Semantics

The SemanticAnalyzer performs:

Pass 1 - Function registration: Build global map function_: (*name* $\rightarrow$ *return_type*, *parameter_types*, *location*). Reject duplicate definitions.

Pass 2 - Per-function checking: Maintain stack of scope maps (scopes_) for block- enter/block-exit, Rules include:

Construct	Rule
Variable declaration	Initializer required; initializer type must match declared type
Variable reference	Must be declared in enclosing scope (inner shadows outer)
Arithmetic (+, -, *, /, %)	Operands: int; result: int
Comparison	Operands: int; result: bool
Logical(&&, , !)	Operands: bool, result: bool
if / while condition	Must be bool
Function call	Callee must exist; arity and argument types must match
return	Expression type must match current function return type

TABLE II: Rules:

Error handling: Panic-mode flag suppresses cascading errors after the first error in a function. Messages format as: semantic error at file:line:col:message.

Known limitation: Reassignment (*x* = *expr*; without type) is not supported-only type name = *expr*; declarations. This simplifies semantic analysis for the initial research prototype.

IV. SYSTEM ARCHITECTURE

A. Pipeline Overview

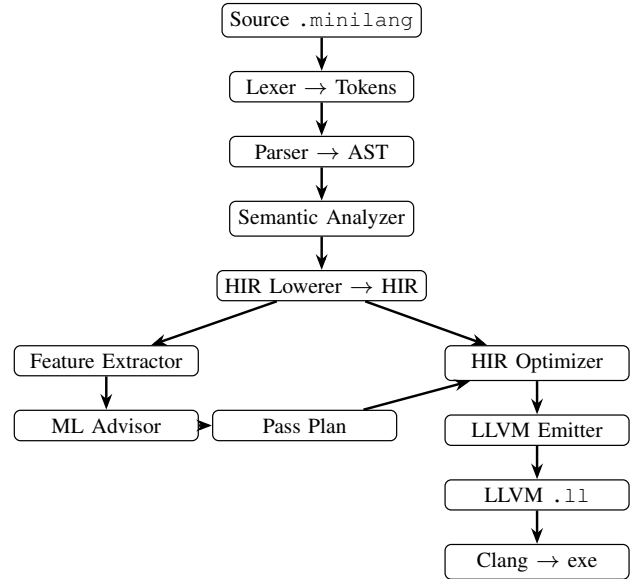


Fig. 1: Data flow.

B. Design Decisions

MiniLang is implemented in C++17 because it aligns with the LLVM ecosystem and is widely used in compiler courses and research. We use a hand-written lexer and recursive-descent parser instead of Flex/Bison so each compilation phase remains easy to read, debug, and explain in a research paper. The following table shows the key design choices.

Decision	Alternatives Considered	Rationale
C++17	Rust, Java	Native LLVM ecosystem; familiar in compiler courses [1]
Hand-written lexer/parser	Flex/Bison, ANTLR	Debuggability; direct mapping to textbook algorithms
HIR before LLVM	Direct AST → LLVM	Isolates front-end optimizations; simpler feature extraction
Textual LLVM IR	LLVM C++ API	Human-readable artifacts; faster research iteration
Separate drivers per phase	Single MiniLang CLI	Independent testing and paper screenshots per stage

TABLE III: Key Design Choices

V. PHASE 1: LEXICAL ANALYSIS

A. Scanner Architecture

The Lexer class maintains:

- `source_`, `filename_` - input buffer and path
- `current_`, `start_`, - scan pointers
- `line_`, `column_` - position tracking
- `had_error` - error flag

Algorithm - Tokenization loop

- `skip_whitespace_and_comments()`
- If at end - return EOF token
- Record token start location
- If letter or '-' - `scan_identifier()` - keyword or Identifier
- If digit - `scan_number()` - `IntLiteral` with `int_value`
- Else switch on punctuation with lookahead for multi-char ops
- On unknown char - Invalid token + error message

B. Comment and Whitespace Handling

`skip_whitespace_and_comments()` loops until non-skippable input:

- Whitespace: advance without emitting tokens
- `//`: consume until newline
- `/*`: consume until `*/` (non-nested)

C. Token Representation

Each Token contains:

- `TokenKind` (enum: `KwInt`, `Plus`, `EqEq`,...)
- `lexeme` (source spelling)
- `SourceLocation`
- optional `<int64_t> int_value` for integer literals

This design avoids re-parsing lexemes in the parser for numeric values.

D. Error Reporting

Invalid characters produce `TokenKind::Invalid` with descriptive messages. The lexer driver sets `had_error_`; compilation aborts with exit code 2 if lexer or parser errors occurred.

VI. PHASE 2: SYNTACTIC ANALYSIS

A. Parser Strategy

The Parser uses recursive descent: each grammar production maps to a `parse_*()` method. The parser holds `current_token` and provides:

- `match(kind)` - consume if current token matches.
- `consume(kind, message)` - require token or error
- `synchronize()` — Panic-mode recovery at statement boundaries (`;`, `}`, keywords)

B. Expression Parsing

Expression parsing follows the precedence chain: `parse_expression()` → `parse_logical_or()` → ... → `parse_primary()` Each level handles left-associative binary operators by looping:

```
// Conceptual structure for parse_term()
left = parse_factor();
while (match(Plus) || match(Minus)) {
    op = previous();
    right = parse_factor();
    left = make_binary(op, left, right);
}
return left;
```

Function calls use `finish_call()`: after parsing identifier `foo`, if (follows, parse argument list.

C. Statement and Function Parsing

- `parse_function()` - type, name, parameters, block body
- `parse_statement()` - dispatches on keyword or type lookahead
- `parse_var_decl()` - requires `=` and expression initializer
- `parse_if_statement()`, `parse_while_statement()` - requires parenthesized condition

D. Parser Output

Successful parse yields Program AST. `dump_program()` provides indented tree output for debugging and paper listings.

E. L^AT_EX-Specific Advice

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

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- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
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- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
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- There is no period after the "et" in the Latin abbreviation "et al."
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [7].

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Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

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TABLE IV: Table Type Styles

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.



Fig. 2: Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an

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ACKNOWLEDGMENT

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Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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